

Assignment 15 (Spring 2025)

Problem 1

Solve the radial wave equation on a circular membrane with $c = 1$, $R = 1$, zero initial velocity, and initial displacement $u(r, 0) = f(r)$, where

$$f(r) = 1.108J_0(2.4048r) - 0.140J_0(5.5201r) + 0.045J_0(8.6537r)$$

Additionally, find all the harmonic frequencies and the radius of each of the (inner) nodal circle for each mode (m) of the solution.

Problem 2

 Solution of 2D wave equation in a circular boundary configuration in a circular membrane, provided that for $u(r, \theta, t)$ is independent of θ will be the following:

$$u(r, t) = \sum_{m=1}^{\infty} W_m(r)G_m(t) = \sum_{m=1}^{\infty} (A_m \cos \lambda_m t + B_m \sin \lambda_m t) J_0\left(\frac{\alpha_m}{R} r\right)$$

where R is the radius of the circle and α_m , shown below, are the roots of the Bessel function $J_0(x)$:

$$\alpha_1 = 2.4048, \quad \alpha_2 = 5.5201, \quad \alpha_3 = 8.6537, \quad \alpha_4 = 11.7915, \quad \alpha_5 = 14.9309$$

and

$$\lambda_m = ck_m = c\alpha_m/R$$

Let's assume $R=0.5$, and $c = 10$.

If initial deflection of the membrane at $t = 0$ is:

$$f(r) = 0.4 J_0(11.0402r) - 0.1 J_0(23.5830r)$$

and initial velocity of the membrane: u_t is 0 at $t = 0$,

- find A_m and B_m for $m=1, 2, 3, 4, 5$
- write all existing terms of $u(r, t)$
- find frequency of all existing harmonics in Hz
- draw the nodal circles for all harmonics and identify the radius of each circle

Problem 3

Find the deflection $u(x, y, t)$ of the square membrane of side π and $c^2 = 1$ for initial velocity 0 and initial deflection

11. $0.1 \sin 2x \sin 4y$

12. $0.01 \sin x \sin y$

13. $0.1 xy(\pi - x)(\pi - y)$

Problem 4

Deflection. Find the deflection of the membrane of sides a and b with $c^2 = 1$ for the initial deflection

$$f(x, y) = \sin \frac{6\pi x}{a} \sin \frac{2\pi y}{b} \text{ and initial velocity } 0.$$

For Reference:

m	α_m
1	2.40483
2	5.52008
3	8.65373
4	11.79153
5	14.93092
6	18.07106
7	21.21164
8	24.35247
9	27.49348
10	30.63461